



Interacting with augmented paper maps: a user experience study

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ABSTRACT

This paper presents a user experience study of interaction with printed maps for providing digitally augmented tourism information. The Interactive Maps system has been implemented based on an interactive printed matter framework which provides all the necessary components for developing smart applications that offer printed matter interaction, and has been deployed and evaluated in the context of the publicly available Tourism InfoPoint of the Municipality of Heraklion. The results of the evaluation highlight that interacting with digitally augmented paper is quite easy and natural, while the overall user experience is positive.

CCS CONCEPTS

- **Human-centered computing** → Empirical studies in HCI
- **Human-centered computing** → Mixed / augmented reality
- **Human-centered computing** → Ambient intelligence

KEYWORDS

Interactive printed matter, interactive paper, natural interaction, ambient intelligence, tabletop interactive system, user experience evaluation

ACM Reference format:

G. Margetis, S. Ntoa, M. Antona, and C. Stephanidis. 2017. Interacting with augmented paper maps: a user experience study. In *Proceedings of 12th Biannual Conference of the Italian SIGCHI Chapter*, Cagliari, Italy, September 2017 (CHIItaly '17), 9 pages. <https://doi.org/10.1145/3125571.3125584>

1. INTRODUCTION

In the last decade, due to the radical evolvement of technology and in particular of portable devices, the idea of a paperless world started to seem very appealing. However, as Sellen and Harper discuss in [28], this idea seems to constitute rather a myth than a tangible reality. On the other hand, new practices of everyday activities related to physical paper (e.g., reading) emerge today, which are totally interlinked with technology [20]. For example, as indicated in [3], a survey done of over 1000 U.S. and 500 U.K. consumers asked their opinions on a variety of issues surrounding paper-based vs. digital media. A sizeable majority of 88% of respondents indicated that they understand, retain and use information better when they read print on paper, while 81% of respondents preferred to read print on paper when given the choice.

Today's technology allows to combine the two worlds, the physical and the digital, potentially retaining the benefits of both. In this context, physical paper augmentation with the use of digital technologies becomes a prominent issue. This paper introduces Interactive Maps, a smart tabletop system that enables interaction with printed maps in order to acquire useful digital tourist information related to points of interest. Information is augmented either upon or laterally to the printed map. Concurrently, the users are still able to take annotations on the printed map and take the map with them afterwards, maintaining unaffected the basic use of the map.

The system was implemented using the innovative interactive printed matter framework discussed in [18]. The framework aims to provide a stepping stone for bridging the use of physical paper with the state-of-the-art technologies that characterize smart environments.

Interactive Maps has been installed at the Tourism InfoPoint of the Municipality of Heraklion, in the island of Crete, Greece¹. The InfoPoint is an original facility by international standards that promotes the island's attractions and provides visitors with novel ways to access information. This paper focuses on an in-situ observation experiment with visitors of the InfoPoint that was carried out with the aim to assess the ease of use of Interactive Maps, the naturalness of interaction with physical paper, as well as the overall user experience.

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CHIItaly '17, September 18–20, 2017, Cagliari, Italy

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ACM ISBN 978-1-4503-5237-6/17/09...\$15.00

<https://doi.org/10.1145/3125571.3125584>

2. THE ERA OF INTERACTIVE PAPER

The idea of implementing a technological environment that merges the everyday use of physical paper with digital information, through natural interaction, is rather old. For example, pioneer research efforts towards this end are DigitalDesk [32] and its successor EnhancedDesk [12], which performed physical paper augmentation offering interaction via touch.

More recent efforts towards integrating digital information in daily activities using physical paper as the basic means of interaction are CoScribe [29], DocuDesk [5], Fact [16], LivePaper [25], PaperWindows [10], PapierCraft [15] and Projective Augmented Books [4]. Some of those systems are tabletops while others are not. A common feature to all of them is that they provide digital augmentation of physical paper through natural interaction.

A first level of categorization of natural interaction with printed matter can be based on whether the user uses their bare hands or other objects (e.g., a stylus) to interact. In [34], five types of finger based interactions with printed matter are defined: (i) interactions based on action time (e.g., short time corresponds to a click, longer time to a double click), (ii) interactions based on hand shapes (e.g., hand fist corresponds to stop or abort, a “V” sign stands for zooming), (iii) interactions based on drawing (e.g., a triangle drawn with a finger might restore an annotation to initial state), (iv) interactions based on movement (e.g., moving hand in the “V” sign upwards stands for zoom in the whole view), and (v) interactions with two hands (e.g., using the two index fingers together doing a pinch gesture indicates the resize of an annotation projected near the printed matter). Holman et al. [10] introduce interaction styles that are performed using digitally augmented paper in the 3D space. They define eight interaction styles based on the natural manipulation of paper: hold, collocate, collate, flip, rub, staple, point and two handed pointing.

On the other hand, the use of stylus or other types of pointers for printed matter interaction supports a reduced set of gestures mainly based on handwriting techniques for annotating on printed matter. In [15] a set of five fundamental types of scope selects are described, inspired by typical marks found on manuscripts, namely underline, margin bar, lasso, crop marks and stitching marks (across two documents). The use of stylus based interaction also incorporates a basic props that hand – based interaction cannot sufficiently compensate, namely handwriting. As mentioned in [31], styli should be used for writing and fingers for drawing, since the outcome of the performed survey highlighted that in terms of legibility and usability using a stylus was better than using a finger to write.

Regarding the digital augmentation paradigms used in interactive paper approaches, Augmented and Mixed Reality are the most prevalent. To this end, various AR approaches emerged.

A common way for creating AR environments over printed matter is through the use of portable devices (PDAs, smart phones, tablets, etc.). For example, the gestured-based system PACER [17] supports fine-grained paper document content manipulation through a smartphone with a touch screen and camera.

Other approaches are based on Head Mount Devices (HMD). HMDs offer greater immersion than portable devices, but they are expensive and cumbersome. Pioneer work towards this direction is MagicBook [1]. The MagicBook system uses a physical book as the main means of interaction. Users look at the pages through an augmented reality display (HMD) and they can observe 3D virtual models appearing out of the pages. A similar approach is discussed in [33], where five educational AR exhibits are presented, based on printed matter containing special markers. In order for the exhibits to provide digital content on top of the printed matter the users wear HMDs.

A second approach to printed matter augmentation has been introduced by Wellner in [32]. This approach requires a projector and a camera lying over the printed matter. While the user interacts, digital content is projected either on top or laterally to the printed matter. The content that is projected is 2D multimedia info or mini interactive applications (e.g., calculator, map view, slide show, etc.). A later approach to augmenting printed books is discussed in [4], where the concept of Projective Augmented Books is proposed. The described system consists of a mobile device with an embedded pico projector and a digital pen. The mobile device’s pico projector is used as a reading lamp in order to project dynamic content onto a book printed on digital paper based on Anoto Technology [23].

A third approach to printed matter augmentation uses a companion display to show the augmented content. For example, Bridging Book [7] is a mixed - media that combines a physical book containing magnets with a touchscreen tablet that incorporates a built-in digital compass. Another example is discussed in [13], which offers physical interaction through printed cards on a tabletop setup, where a simple webcam monitors the table's surface and identifies the thrown cards. The system consists of two mini games based on the printed cards shown on the table: a multiple-choice quiz and a geography-related game.

Despite the many research efforts discussed above, a significant gap today is the lack of a systematic approach to facilitate the development and integration of interactive paper applications in smart environments. A recent approach towards this direction is the interactive paper framework introduced in [18], which provides a means for implementing interactive paper applications and systems that can be incorporated in smart environments. This framework, whose basic features are discussed briefly in the next section, has been used to implement the Interactive Maps system.

¹ <http://ami.ics.forth.gr/installation/info-point/>

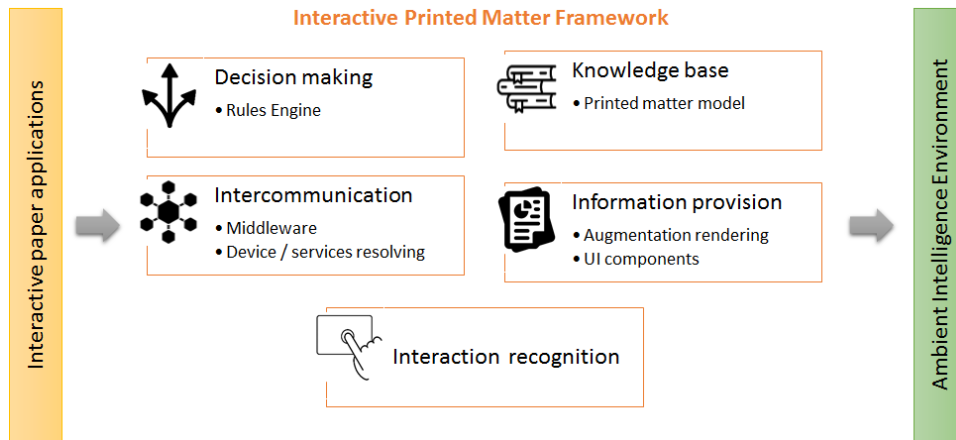


Figure 1: The Interactive Printed Matter Framework.

3. THE INTERACTIVE PRINTED MATTER FRAMEWORK

The Interactive Printed Matter framework provides all the necessary components to facilitate the integration of applications that use printed matter interaction in smart environments. The framework has been implemented according to a service-oriented-architecture (SoA), which makes its integration into already existing smart environments easy. An ontology-based model for printed matter interaction modelling in smart environments has been defined, which can be easily extended in order for the framework to readily incorporate new means of information acquisition, processing and visualization that may emerge in the future, thus making it fully extensible.

Figure 1 illustrates the essential components of the framework. In more details, a meta-ontology model that represents interaction with printed matter in smart environments has been specified, constituting the stepping stone for the knowledge base materialization that corresponds to each smart environment. The printed matter model defines the relationships and the entities that describe smart environments, user profiles, content representation, and most importantly printed matter interaction and augmentation.

Furthermore, the decision making component provides context-aware decisions regarding the selection of the appropriate UI components and corresponding content, according to the user's interaction. This component is also responsible for keeping the current state of the smart environment, populating it with necessary information regarding the available systems, services and devices that have been installed and operate in it.

The framework intercommunication functionality is based on FAmINE [9], a middleware providing the necessary seamless interoperability of the devices and services that comprise the Aml ecosystem.

Regarding interaction management, the framework incorporates a dedicated component which is aware of all the user

interactions performed in a smart space and related to printed matter manipulation. This particular component accepts input from all the available I/O devices/agents/services of the ecosystem and correlates it with recognized printed matter.

The interaction techniques that are currently supported are: handwriting, gestures, touch / click, printed matter direct manipulation (i.e., a specific piece appeared, moved or disappeared with regard to a system) and combined interaction (i.e., when a user combines two or more pieces of printed matter in order to accomplish more complex tasks).

Information provision is supported through digital augmentation either upon or laterally to the printed matter. The supported augmentations can be used directly by the applications through an API provided by the framework. Each application developed using the proposed framework is able to: (a) listen for augmentation results that stem after a user's interaction and reasoning processing, (b) acquire digital assets, in a proper format, related to the augmentation result, and (c) render this information to a system (using a UI toolkit which is also included as a complementary library in the framework).

In summary, the framework provides: (a) an open architecture enabling integration of new types of technologies for information acquisition and provision (b) independency of the development technologies for the applications (c) an extensible ontology based reference model for printed matter, as well as context-awareness mechanisms (d) implementation of printed matter augmentation mechanisms in the environment (e) support for multimodal natural interaction with printed matter in smart environments.

4. THE INTERACTIVE MAPS SYSTEM

The Interactive Maps system is an adapted version of the Interactive Documents system [19], targeted to facilitate interaction with printed maps as well as their augmentation with digital information. More specifically, it is a tabletop system on which visitors may place a printed map of the historical center of Heraklion, freely distributed at the InfoPoint (Figure 2).



Figure 2: LEFT: The Interactive Maps system installed at the InfoPoint, Tourism Office, Municipality of Heraklion. RIGHT: a user selects through touch a hotspot that is superimposed on the printed map. The system acquires related information (e.g., images and text) and displays it laterally to the map.

Once the map is placed on the table, the system annotates the hotspots that are digitally augmented and thus become interactive, assisting users in easily identifying points of information on the printed map. Users can touch on the printed map hotspots in order to receive further information, which is presented laterally to the map. More specifically, for each hotspot, multimedia in several languages is made available to the user, including textual information, images and videos. As a result, the city visitors can have a preview of several sightseeing points and plan in advance their tour in the historical center of the city. At the same time, the printed map can be used in a traditional manner, for example by annotating points of interest or routes. Finally, visitors can take the printed map with them to guide them in their city tour.

Interactive Maps comprises a desk, over which a projector, a high resolution camera and a depth sensor have been installed. The system is integrated into a regular PC (i.e., Intel i7 CPU, 8GB RAM). The camera overlooks what is happening on the desk's surface and feeds real time the application with captured images. The application is able to recognize predefined printed maps and track them continuously while they are placed on the desk's surface. The depth sensor is used in order to provide depth images of objects that are placed over or near the surface. These images are analyzed in order for the system to be able to recognize and eventually track human fingers. Every time a finger touches a map on the desk, the application examines if this touch should trigger a hotspot trigger event. In such case, the system searches and acquires digital assets that adhere to the corresponding hotspot of the map. Then these assets are compiled, formatted and displayed laterally to the map nearby the engaged hotspot [19].

For the implementation of the Interactive Maps system the aforementioned Interactive Printed Matter framework has been used. To this end, the development of such a system can be mainly focused on application intelligence letting the framework

undertake any basic functionality related to interaction with physical printed matter and to its augmentation.

5. EVALUATION

The system has been evaluated in-situ for five days with actual visitors of the Heraklion InfoPoint during its official working hours. Typically, the InfoPoint has in average 80 visitors per day during the high tourist season (from early spring until late autumn). This evaluation was part of the overall user experience (UX) evaluation of seven interactive systems that have been installed at the InfoPoint, the results of which will be reported elsewhere. The evaluation of Interactive Maps aimed at assessing the overall UX with the system focusing on interaction with digitally augmented physical paper, and in more details to explore the following hypotheses:

H1. The system is easy to use with minimum guidance.

H2. Interaction with paper is natural.

H3. The overall user experience is positive.

Other hypotheses regarding cross-relations of the UX with the participants' age and computer expertise were also considered, but the results will be discussed elsewhere.

5.1 Methodology

In situ evaluation promises benefits associated with exploring how a system is actually used in its real environment, avoiding contrived situations imposed in laboratory and field testing [6]. For instance, an important benefit of in-situ studies is that they can reveal how the environment itself can have quite a different impact on the user experience [26]. Although usability evaluations in laboratory settings can identify many usability issues, actual use aspects are less easily revealed in such settings, where participants are asked by an experimenter to follow instructions and perform

certain tasks using a prototype system [2]. Comparisons of in situ and laboratory evaluation, carrying out evaluations in the exact same way, indicated that more usability problems were identified in situ, while it was only this type of evaluation that revealed problems related to cognitive load and interaction style [22]. An important parameter that may affect the overall evaluation results is users' motivation to use the system in the real world, which is partially determined by users' curiosity and system's novelty [24].

Despite the high potential of in-situ (or in-the-wild) evaluations, there are several challenges associated, as the high cost involved and the high expectations from such studies regarding the anticipated outcomes [2]. Other considerable challenges include the lack of control, as well as environmental conditions, or social considerations [11]. Furthermore, an important challenge to be addressed is how and what will be recorded. In such situations when it is impossible, yet also not desirable, to capture everything, the key is to use a combination of various methods that reveal both hoped for and unexpected effects of the context of use [26], [27]. It has also been suggested as a best practice in such settings to supplement objective context data with subjective data on participants' context perceptions [27]. Finally, a significant concern for any observation experiment is whether to use any video, photo, or audio recording. Especially in the case of in situ evaluations, it is of uttermost importance to respect social norms and switch to other recording methods, such as note taking, if multimedia recording is considered socially or legally unacceptable [27].

Before determining the exact methodology and tools, it was important to fully understand the context. Therefore, the research team delved deep into the InfoPoint daily routines and visitors' behavior around the interactive systems by observing and documenting points of interest and points of caution. These observations along with state of the art research in the field dictated the need for an evaluation combining objective and subjective assessments, namely observation, semi-structured interviews with users, semi-structured interviews with employees of the InfoPoint, as well as questionnaires filled-in by the users themselves.

Observations

Although video recording in public spaces is allowed upon approval of the Hellenic Data Protection Authority, given the diversity of cultural backgrounds and social norms of the potential target users, it was decided to not use any multimedia recording, and make thus the evaluation process less intrusive. In order to compensate the lack of video recording, free note taking was complemented with a structured observation grid, aiming to record the following parameters based on [21]: appeal, learnability, effects of breakdowns, distraction and further enhanced with parameters related to cultural readiness, flexibility, and interaction behavior as explained in Table 1.

Semi-structured interviews

The semi-structured interviews with users and employees were part of the overall InfoPoint UX evaluation and included also questions that do not pertain specifically to the Interactive Maps

system. Nevertheless, all the responses that refer to the specific system are analyzed in this paper (e.g., if the system was mentioned as one of the most liked/most disliked, or if any specific comment was made regarding it).

Questionnaires

Users were asked to fill-in a questionnaire to rate their experience. The questionnaire included: (i) metrics inspired from [21], [30] provided as bipolar adjective pairs on a 5-point semantic differential scale [8], and (ii) open-ended comments. The adjective pairs were the following: Attractive/Ugly, Enjoyable/Boring, Convenient/Tiresome, Pleasant/Unpleasant, Useful/Useless, Informative/Uninformative, Satisfactory/Unsatisfactory, Easy to use/Difficult to use, Unique/Ordinary.

Table 1: List of metrics of the observation sheet

Parameter	Metrics
Appeal	User's initial reaction to the display (e.g., curious, excited, confused, stressed, etc.)
Learnability	Instructions read, understanding how to use the display by oneself, assistance requests, difficulties encountered
Breakdown effects	Technical problems, problems overcome by the user
Attention and distractions	Functionality preferred, distractions
Cultural readiness	Users' comfortability with the display, unexpected actions
Flexibility	Accomplishment of tasks not originally envisioned
Interaction behavior	User's interaction behavior (e.g., confidence, autonomy, insecurity, impatience, etc.), functionality used

5.2 Procedure and Participants

Visitors were informed of the evaluation and the procedures that would be followed upon approaching the first interactive system installed at the InfoPoint premises. Those who orally agreed to be observed during their interaction with the systems were given an informed consent form to sign. As the interactive systems are installed adjacently at the InfoPoint, one dedicated observer for every two systems had been assigned for recording user interactions using the observation sheets. Once visitors completed their interaction with the InfoPoint systems and/or with the staff for retrieving information, and before leaving the building, they were given the questionnaires to fill-in (one for each system they had interacted with), and were interviewed.

In total, the Interactive Maps system was used by 87 visitors, individually, in pairs or small groups of three or four visitors, resulting in 46 system usages (individual: 15, pairs: 24, small

groups: 7). Participant ages ranged from children to seniors, as follows: 5 children (younger than 12 years old), 1 teenager (12-17 years old), 24 youth (17-30 years old), 35 young adults (30-44 years old), 21 adults (44-60 years old), 1 senior (older than 60 years).

The first part of the questionnaire included additional demographic information, namely gender, self-reported level of computer skills, frequency of computer use, education level, as well as if their studies or profession is related to tourism or computer science. Table 2 presents the participants' demographic information, as acquired from 31 participants who filled out this questionnaire section.

Table 2: Participants' demographic information

Gender		Computer skills	
Male	67.74%	Low	3.23%
Female	32.26%	Intermediate	54.84%
		High	41.94%
Education		Frequency of computer use	
Less than high school	3.23%	Every day	93.55%
High school or equivalent	16.13%	Several times a week	3.23%
College but no degree	16.13%	Several times a month	0.00%
Associate degree	9.68%	Almost never	3.23%
Bachelor degree	32.26%	Never	0.00%
Graduate degree	22.58%		
Tourism-related studies or profession		Computer-related studies or profession	
No	90.32%	No	61.29%
Yes	9.68%	Yes	38.71%

5.3 Results

The initial hypotheses have been explored by combining data recorded in the observation sheets with questionnaire data. This section reports the results for each of the three hypothesis and also insights from the qualitative analysis of the free text comments and interviews with visitors and employees.

H1. The system is easy to use with minimum guidance

In order to study H1, the following observation sheet recordings were used: if the user read the instructions provided by the system, if they understood how to use the display by themselves, if they requested assistance and for what, as well as what difficulties they did encounter during system usage. Also participants'

questionnaire rating for the easy to use/difficult to use adjectives pair are reported.

Figure 3 illustrates the results recorded in the observation sheets for the aforementioned questions. Error bars in the chart represent 95% confidence intervals calculated with the Adjusted Wald method [14]. A correlation of observation data for the first three questions highlighted that, out of the 8 participants who read instructions, 7 (87.5%) understood how to use the system by themselves, while only 1 participant (12.5%) had to ask for assistance in order to understand how to use the system.

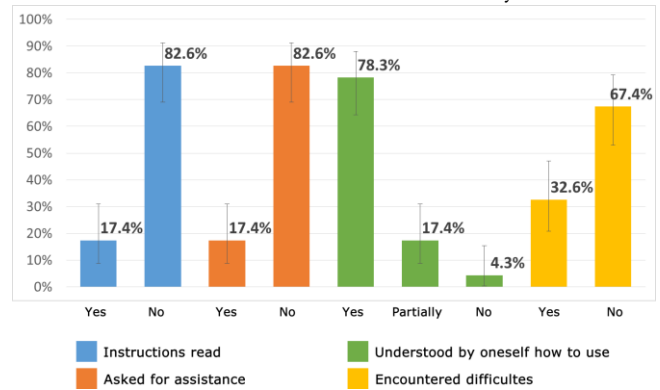


Figure 3: Observation data related to H1

Further analysis of the difficulties encountered revealed that out of the 15 participants who faced difficulties, 12 (80%) encountered problems related to touch, 4 (26.67%) tried to carry out an unsupported gesture – namely swipe and moving a popup, while 1 (6.67%) faced a problem since the printed map was out of the camera recognition bounds. A qualitative analysis of observers' notes suggested that visitors seemed familiar with touch gestures, perhaps having prior experience with smartphones or tablets, and had no specific problems in extending the use of gestures from mobile devices to printed matter. The responsiveness of such devices however is quite more immediate than that of gestures detected through computer vision, a technology which was not visible to participants. As a result, the system did not adhere to their expected response time, leading to repeated touches, which was recorded as a difficulty.

Participants' questionnaire responses regarding ease of use resulted in an average score of 1.09, on a scale from -2 (very difficult to use) to 2 (very easy to use), with the 95% confidence interval ranging from 0.69 to 1.5. Further analysis of the answers revealed that only 1 participant rated the system as very difficult to use (3%, CI[UL]: 17%) and only two participants rated it as difficult to use (6.2%, CI[UL]: 21%).

In summary, it can be concluded that we are 95% confident that for the general population at least 64% will understand by themselves and at least 8% will partially understand by themselves how to use the system, while at least 53% will not encounter any difficulties at all. Furthermore, none of the recorded difficulties constitutes a showstopper, further verified by the fact that none of the participants quitted system usage. Finally, the system was

rated as overall easy to use, while a very small proportion of participants rated it as difficult or very difficult to use.

H2. Interaction with paper is natural.

This hypothesis was explored through the following metrics: functionality used, functionality preferred, tasks carried out that had not been originally envisioned, and unexpected / unusual actions. As functionality preferred, observers recorded the system functionality more often used during a session. It was observed that almost all the participants employed touch to interact with the system ($M=95.65\%$, 95% CI [84.66%, 99.61%]), while approximately half of them engaged in additional activities such as move the physical map on the table ($M=52.17\%$, 95% CI [38.14%, 65.88%]), or browse the gallery of photos ($M=54.35\%$, 95% CI [40.18%, 67.85%]). Furthermore, touching hotspots on the printed map and moving the map on the table surface were the actions mostly preferred (86.96% and 34.78% correspondingly). Given the high proportion of people who touched the digitally annotated hotspots on the printed map, and that we can be 95% confident that at least 84.6% of the general population would carry out this action with the given system, it is evident that interaction with the system (and in particular touch-based interaction) comes natural to most users.

Regarding unexpected / unusual actions, it was observed that 7 users (36.84%) carried out a swipe gesture to browse the media gallery contents. Although the swipe gesture was not supported, the fact that it was expected, further supports the hypothesis that touch-based interactions on digitally augmented paper seemed natural to users. Moreover, 6 users in total extended interaction with the system and complemented it with interaction with the physical paper (e.g., by combining the map on the table surface with a map of their own, or by taking the printed map with them), which is a positive indication towards endorsing convergence of the digital and physical world.

H3. The overall user experience is positive.

This hypothesis was explored by studying the observed users' reactions to the display, their interaction behavior and comfortability with the display, as well as by analyzing any distractions imposed by the system that might disorient users from achieving their current task at hand, and any major technical problems, as well as their effects. The observation data are further complemented with analysis of the questionnaires filled-in by the users.

As illustrated in Figure 4 (left), users' reactions were mainly positive, the most common reaction was that of curiosity (65.22%), which is in accordance with previous research on usage motivators for public display systems [24]. It should be noted that

for each user all the exhibited reactions - positive, or negative - as they approached the system were recorded. For instance, a user might approach the system exhibiting curiosity, while after approaching they might look confused. The recorded reactions have been further classified as positive and negative, according to how they would motivate or demotivate system usage. Analysis of the aggregated data showed that 79.06% of the users had only positive reactions (95% CI [64.58%, 88.79%]), 11.62% only negative (95% CI [4.61%, 24.94%]) and 9.3% mixed reactions (95% CI [3.12%, 22.16%]).

Regarding users' interaction behaviors (Figure 4 right), the most often observed positive ones were autonomy (39.13%), (58.7%) and exploratory disposition (47.83%), while the most common negative reactions were impatience (17.39%) and difficulty (13.04%). Difficulties mostly refer to the rather slow responsiveness of the system to touch gestures, as it has been already analyzed in H1, a system behavior which also caused impatience to users, as well as the majority of the negative recorded attitudes, namely confusion, frustration, and insecurity. Further analysis of the recorded behaviors attempted to identify sessions where only positive ($M=63.04\%$, 95% CI [48.57%, 75.51%]), only negative ($M=8.69\%$, 95% CI [2.9%, 20.86%]) or mixed behaviors ($M=28.26\%$, 95% CI [17.22%, 42.65%]) were recorded.

The analysis of the exhibited comfortability with using the system suggested that the majority of users felt comfortable ($M=69.57\%$, 95% CI [55.11%, 81.00%]) or partially comfortable ($M=10.87\%$, 95% CI [4.29%, 23.49%]), while 19.57% (95% CI [10.43%, 33.39%]) did not feel comfortable.

On the other hand, analysis of the questionnaires showed an overall positive attitude towards the system, as illustrated in Figure 5, where it is evident that negative feelings - represented towards the center of the circle - received very few points. Analysis of the 95% CI revealed that the lowest possible scores CI[LL] for the wide public are 0.69 for ease of use, 0.78 for satisfaction and 0.85 for convenience, which all represent positive attitudes. On the other hand, the results of the semi-structured interviews with users indicated that the Interactive Maps system was often mentioned as one of the users' favorite systems at the InfoPoint ($M=46.87\%$, 95% CI [30.87%, 63.55%]), while only two users said that it was one of the system they disliked most ($M=6.25\%$, 95% CI [0.72%, 21.16%]), one due to its graphic design and the other due to the slow system responsiveness to touch gestures. Finally, interviews with employees of the InfoPoint also confirmed that the Interactive Maps system is one of the most often preferred by visitors who are interested in retrieving information and not only be entertained.

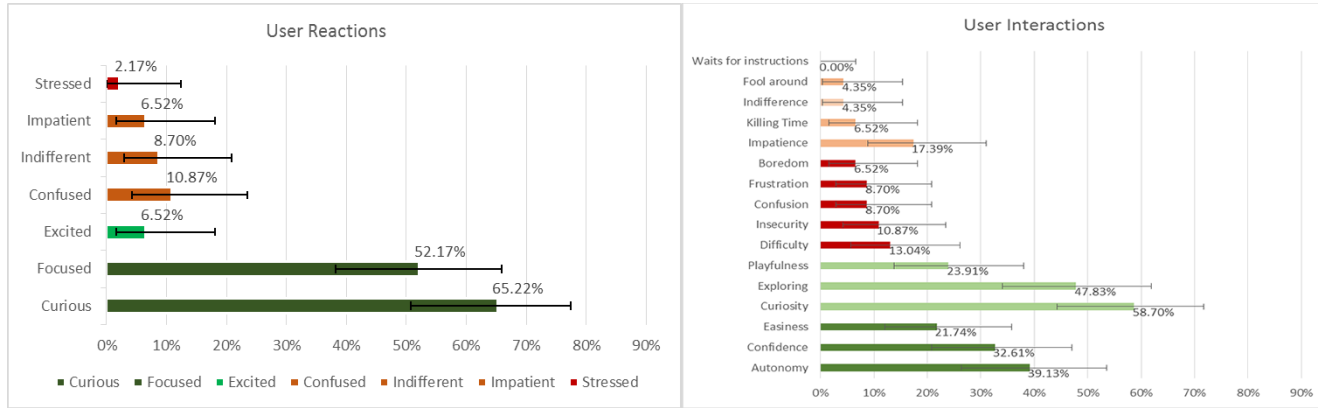


Figure 4: LEFT: User reactions towards the system. RIGHT: User interactions with the system. Error bars illustrate the 95% CI.

In summary, in the context of studying the overall UX the following conclusions have been reached: (i) 79.06% of the users had only positive reactions towards the system and we are 95% certain that at least 64.58% of the wide public will react positively, (ii) 63.04% of interaction behaviors were in total positive and we are 95% certain that at least 48.57% of the general population will exhibit purely positive attitudes, (iii) the majority of users felt comfortable or partially comfortable with using the display and we are 95% confident that at least ~60% of the wide public will feel as such and (iv) the subjective views of the users revealed an overall positive attitude towards the system, while it is 95% certain that the same holds for the wide public, as even the lowest possible scores are greater than 0, which the score for neutral attitude.

Furthermore, although the same individuals participated in the observation and the questionnaire and mini-interview, due to the flow of visitors in the InfoPoint it was not possible to correlate observations to interviews / questionnaires. However, the results have been used independently, yet in synergy where appropriate, in order to explore the different hypotheses.

Finally, lack of video recordings might result in less data. However, skilled observers (UX experts having carried out numerous evaluation experiments) were employed for recording interactions, specific structured forms were used, while each observer was responsible for one or two specific systems becoming thus more efficient at recordings. Furthermore, observation sheets were reviewed and processed on a daily basis, when the experiment was still in memory.

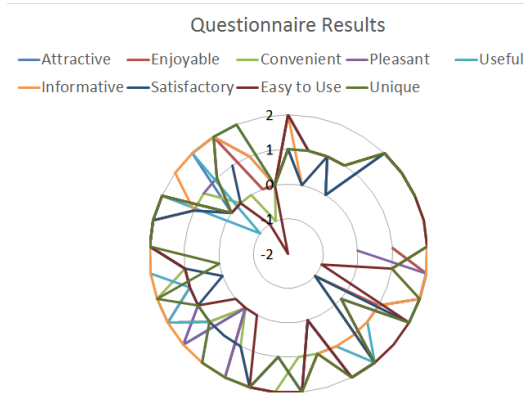


Figure 5: Questionnaire results

5.4 Limitations of the current study

An important consideration refers to the user sample that participated in the study. As the evaluation was an in-situ observation, it was not possible to have a balanced sample in terms of age, gender, education, or computer skills. This is a limitation intrinsic in the selected set up and method.

6. CONCLUSIONS AND FUTURE WORK

This paper has presented Interactive Maps, a system that has been implemented using the Interactive Printed Matter framework, which provides readily augmentation of printed matter and support for multimodal natural interaction with printed matter in smart environments. More specifically, Interactive Maps employs touch-based interaction with maps placed on a plain table, and augments the printed map with digital information related to the map hotspot that has been selected by the user. The system has been installed for public use at the Tourism InfoPoint of the Municipality of Heraklion in Crete, Greece. Taking advantage of the high number of visitors of the InfoPoint, an in-situ evaluation of the system has been carried out involving 46 system usages by visitors, single or in small groups. The goal of the evaluation was to assess interaction with digitally augmented printed matter, exploring its ease of use, naturalness, and overall UX. The results of the evaluation highlight that although users did not receive any guidance prior to the system usage their majority easily understood how to use it. More importantly, interaction with the digital augmented map through touch was natural for users, while they were overall satisfied by the various attributes of the system. Future reports on this evaluation will discuss the effect of user age

and computer expertise on the interaction, user attitude towards the system, and overall UX.

ACKNOWLEDGMENTS

The authors would like to thank Mrs. Ilia Adami, Mrs. Irini Kontaki, Mr. Vasilios Kouroumalis and Dr. Antigoni Parmaxi for their participation in designing and running the UX evaluation of the InfoPoint interactive systems. This work is supported by the FORTH-ICS internal RTD Programme 'Ambient Intelligence and Smart Environments'.

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